Deep Learning: From Multilayer Perceptrons to Transformers

Reading and Reference List

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Contents

Reading and Reference List

Companion to `syllabus.pdf`.

Tiering. [**C**] = core, assigned and examinable. [**S**] = supplementary, assigned to some weeks. [**R**] = reference, for depth or for projects.

Where a paper is best known by a short name, that name appears in brackets.

Assessment pointers. Core entries carry a mark of the form PS2 A2, or Lab 6(b), or midterm, naming where skipping the reading will cost you. The tiering says how hard a reading is being pushed; it never said what happened if you ignored it, which made [C] indistinguishable from [S] in practice, at 2 a.m., in Week 9.

Three things follow from reading those pointers carefully.

A **Lab** pointer means the paper's claim is one *you* will re-measure. It is not enough to know what the paper concluded; you need to know what it measured and under what conditions, because you will be comparing your numbers against it and explaining the gap.

A **midterm** pointer with no accompanying assignment means the reading is examined but never rehearsed. There are six of these and they are the ones students under-prepare — four of them in §2.

Entries with no pointer are context. They are still assigned. They will not be marked.

Nineteen of the entries below are papers whose central claim you will personally re-measure at reduced scale. In four cases — Table 3 of Vaswani, the Chinchilla ratio, BYOL's collapse resistance, and the Kaplan exponents — **your measurement is expected to disagree with the published one**, and the assessment asks you to say why rather than to make it agree. Those four are collected in §16.

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0. Books and long-form references

[**C**] **Prince, S. J. D.** (2023). *Understanding Deep Learning.* MIT Press. — The primary text. Free online. Best current balance of mathematical care and readability; the figures are unusually good. Chapters map cleanly onto Weeks 1–7.

[**C**] **Goodfellow, I., Bengio, Y., & Courville, A.** (2016). *Deep Learning.* MIT Press. — Canonical reference. Parts I–II remain the standard treatment of the foundations; Part III has aged. Chapter 11 (practical methodology) is assigned in Week 13.

[S] Jurafsky, D., & Martin, J. H. *Speech and Language Processing*, 3rd ed. draft. — Ch. 2 (tokenization), Ch. 9 (the Transformer), Ch. 10 (masked language models). Continuously updated; check the date on the chapter you download.

[S] Torralba, A., Isola, P., & Freeman, W. T. (2024). *Foundations of Computer Vision.* MIT Press. — Convolution and representation-learning chapters. Free online.

[S] Bishop, C. M., & Bishop, H. (2024). *Deep Learning: Foundations and Concepts.* Springer. — Probabilistic framing throughout. Preferable for students coming from a statistics background.

[R] Murphy, K. P. (2022, 2023). *Probabilistic Machine Learning: An Introduction* and *Advanced Topics.* MIT Press. — Encyclopedic. Use as a lookup reference rather than a read-through.

[R] Zhang, A., Lipton, Z., Li, M., & Smola, A. *Dive into Deep Learning.* — Every derivation accompanied by runnable code. Useful when an implementation detail is unclear.

[R] Roberts, D. A., Yaida, S., & Hanin, B. (2022). *The Principles of Deep Learning Theory.* Cambridge. — Effective-theory treatment of initialization and finite-width corrections. Demanding but the most systematic account of signal propagation available.

[R] Telgarsky, M. *Deep Learning Theory* lecture notes. — §2 and §5 assigned in Week 3. Written at research level; extract the intuitions on a first pass.

[R] Griewank, A., & Walther, A. (2008). *Evaluating Derivatives*, 2nd ed. SIAM. — The definitive treatment of automatic differentiation. Consult if PS1 raises questions the survey does not answer; the checkpointing chapter is where the $\sqrt{k}$ result comes from.

[R] Hamilton, W. L. (2020). *Graph Representation Learning.* — Reference for the GNN portion of Week 7's unification lecture.

[R] Nielsen, M. *Neural Networks and Deep Learning.* — Free, intuition-first. Recommended only for students who found Week 2 hard.

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1. Origins and foundations

[C] Rosenblatt, F. (1958). The perceptron: a probabilistic model for information storage and organization in the brain. *Psychological Review*, 65(6). — Read for the framing, not the algorithm.

[S] Minsky, M., & Papert, S. (1969). *Perceptrons.* MIT Press. Ch. 1–2. — The XOR obstruction and the argument that stalled the field.

[C] Rumelhart, D. E., Hinton, G. E., & Williams, R. J. (1986). Learning representations by back-propagating errors. *Nature*, 323. — Week 2. Note carefully what was new in 1986 and what was not: reverse-mode AD predates it by decades in other fields. midterm

[C] Baydin, A. G., Pearlmutter, B. A., Radul, A. A., & Siskind, J. M. (2018). Automatic differentiation in machine learning: a survey. *JMLR*, 18. — The clearest statement of forward versus reverse mode and the cost argument. §22 item 3

[S] **LeCun, Y., Bottou, L., Orr, G., & Müller, K.-R.** (1998). Efficient backprop. In *Neural Networks: Tricks of the Trade.* — Much of the modern initialization and preprocessing folklore, twelve years early.

[S] **Karpathy, A.** (2016). Yes you should understand backprop. Blog. — Short. Assigned in Week 2 because the failure modes it lists are exactly the six planted in Lab 2. Read it before the session, not after. **Lab 2 (all)**

[R] **Werbos, P.** (1974). *Beyond Regression.* PhD thesis, Harvard. — Prior discovery of backpropagation. Historical.

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2. Approximation theory and expressivity

[C] **Cybenko, G.** (1989). Approximation by superpositions of a sigmoidal function. *Mathematics of Control, Signals and Systems*, 2(4). — Read for the precise statement. Density in $C(K)$; no width bound; no learnability claim. §22 item 6

[C] **Hornik, K.** (1991). Approximation capabilities of multilayer feedforward networks. *Neural Networks*, 4(2). — Generalizes Cybenko beyond sigmoids. §22 item 6

[C] **Barron, A. R.** (1993). Universal approximation bounds for superpositions of a sigmoidal function. *IEEE Transactions on Information Theory*, 39(3). — The result that actually matters: $O(1/\sqrt{m})$ rate independent of input dimension for functions with bounded Fourier moment. This is what breaks the curse of dimensionality, and it is why “universal approximation” alone is nearly vacuous. §22 item 7

[C] **Telgarsky, M.** (2016). Benefits of depth in neural networks. *COLT*. — The sawtooth construction: a function computable by a depth-k network requiring exponential width at depth $O(\sqrt{k})$. §22 item 8

[S] **Eldan, R., & Shamir, O.** (2016). The power of depth for feedforward neural networks. *COLT*. — Depth separation for a radial function; complementary construction to Telgarsky’s.

[R] **Hanin, B., & Rolnick, D.** (2019). Complexity of linear regions in deep networks. *ICML*. — Corrects the widespread overestimate of how many linear regions trained ReLU networks actually use.

[R] **Poole, B., Lahiri, S., Raghu, M., Sohl-Dickstein, J., & Ganguli, S.** (2016). Exponential expressivity in deep neural networks through transient chaos. *NeurIPS*.

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3. Optimization

[C] **Kingma, D. P., & Ba, J.** (2015). Adam: a method for stochastic optimization. *ICLR*. — Week 1. §22 item 30

[C] **Loshchilov, I., & Hutter, F.** (2019). Decoupled weight decay regularization. *ICLR*. [AdamW] — Why L_2 penalty and weight decay are not the same thing under adaptive methods.

§22 item 31

[C] Reddi, S. J., Kale, S., & Kumar, S. (2018). On the convergence of Adam and beyond. *ICLR*. — The error in Adam's original convergence proof. Assigned partly as a lesson about trusting published theory.

[C] Yang, G., & Hu, E. J. (2021). Tensor Programs IV: feature learning in infinite-width neural networks. *ICML*. — The feature-learning versus lazy-regime distinction that motivates μP. PS4 A3

[C] Yang, G., Hu, E. J., Babuschkin, I., et al. (2022). Tensor Programs V: tuning large neural networks via zero-shot hyperparameter transfer. *NeurIPS*. [μP] — Week 9. The scaling table is the practical takeaway; the derivation is worth the effort. PS4 A2, **Lab 9**

[S] Duchi, J., Hazan, E., & Singer, Y. (2011). Adaptive subgradient methods. *JMLR*, 12. [AdaGrad]

[S] Polyak, B. T. (1964). Some methods of speeding up the convergence of iteration methods. *USSR Comp. Math.* — Momentum, in the original.

[S] Nesterov, Y. (1983). A method for solving the convex programming problem with convergence rate $O(1/k^2)$.

[S] Goyal, P., Dollár, P., Girshick, R., et al. (2017). Accurate, large minibatch SGD: training ImageNet in 1 hour. — The linear scaling rule and gradual warmup.

[S] McCandlish, S., Kaplan, J., Amodei, D., et al. (2018). An empirical model of large-batch training. — Gradient noise scale and critical batch size.

[S] Loshchilov, I., & Hutter, F. (2017). SGDR: stochastic gradient descent with warm restarts. *ICLR*. — Cosine schedules.

[R] Keskar, N. S., Mudigere, D., Nocedal, J., et al. (2017). On large-batch training for deep learning: generalization gap and sharp minima. *ICLR*.

[R] Li, H., Xu, Z., Taylor, G., Studer, C., & Goldstein, T. (2018). Visualizing the loss landscape of neural nets. *NeurIPS*. — Source of the filter-normalized landscape plots; evidence for the residual-smoothing claim.

[R] Shazeer, N., & Stern, M. (2018). Adafactor: adaptive learning rates with sublinear memory cost. *ICML*.

[R] Gupta, V., Koren, T., & Singer, Y. (2018). Shampoo: preconditioned stochastic tensor optimization. *ICML*.

[R] Bernstein, J., & Newhouse, L. (2024). Modular duality in deep learning. — The norm-selection view of optimizer design; the theoretical frame behind Muon and related methods.

[R] Goh, G. (2017). Why momentum really works. *Distill*. — Interactive; the best intuition-builder available for momentum.

4. Initialization, normalization, activations

[C] Glorot, X., & Bengio, Y. (2010). Understanding the difficulty of training deep feedforward neural networks. *AISTATS*. [Xavier init] — Derive the constant yourself; do not memorize it. §22 item 10, **Lab 3**

[C] He, K., Zhang, X., Ren, S., & Sun, J. (2015). Delving deep into rectifiers. *ICCV*. [He init, PReLU] — The factor-of-two correction for ReLU. §22 items 9 and 11, **Lab 3 (all)**

[C] Ioffe, S., & Szegedy, C. (2015). Batch normalization. *ICML*. PS2 A2, **B2**

[C] Santurkar, S., Tsipras, D., Ilyas, A., & Madry, A. (2018). How does batch normalization help optimization? *NeurIPS*. — Assign with the above. The internal-covariate-shift explanation does not survive the experiments here, and the pairing is a useful lesson in how techniques outlive their justifications. PS2 A5

[C] Ba, J. L., Kiros, J. R., & Hinton, G. E. (2016). Layer normalization. — Why sequence models cannot use batch statistics. PS2 B4

[C] Xiong, R., Yang, Y., He, D., et al. (2020). On layer normalization in the Transformer architecture. *ICML*. — Pre-LN versus post-LN, and the explanation for why the original recipe needed warmup. §22 item 25, **Lab 7(b,c)**

[S] Nair, V., & Hinton, G. E. (2010). Rectified linear units improve restricted Boltzmann machines. *ICML*.

[S] Zhang, B., & Sennrich, R. (2019). Root mean square layer normalization. *NeurIPS*. [RMSNorm] — Now standard in large models.

[S] Saxe, A. M., McClelland, J. L., & Ganguli, S. (2014). Exact solutions to the nonlinear dynamics of learning in deep linear networks. *ICLR*. — Orthogonal initialization; dynamical isometry.

[S] Hendrycks, D., & Gimpel, K. (2016). Gaussian error linear units. [GELU]

[S] Shazeer, N. (2020). GLU variants improve Transformer. — SwiGLU, now the default FFN in most large models.

[R] Wu, Y., & He, K. (2018). Group normalization. *ECCV*.

[R] Ulyanov, D., Vedaldi, A., & Lempitsky, V. (2016). Instance normalization.

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5. Regularization and generalization

[C] Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., & Salakhutdinov, R. (2014). Dropout. *JMLR*, 15.

[C] Zhang, C., Bengio, S., Hardt, M., Recht, B., & Vinyals, O. (2017). Understanding deep learning requires rethinking generalization. *ICLR*. — Week 8. The random-label experiment. **Lab 8 (all)**

[C] Belkin, M., Hsu, D., Ma, S., & Mandal, S. (2019). Reconciling modern machine learning practice and the classical bias–variance trade-off. *PNAS*, 116(32). — Double descent. **Lab 8(b)**

[C] Nakkiran, P., Kaplan, J., Bansal, Y., et al. (2020). Deep double descent: where bigger models and more data hurt. *ICLR.* — Extends double descent to model size, data size, and training time. **Lab 8(b)**

[C] Wilson, A. G. (2025). Deep learning is not so mysterious or different. — Assign against Zhang et al. The disagreement about whether deep generalization requires new theory is live and students should see both positions argued well. midterm

[C] Jacot, A., Gabriel, F., & Hongler, C. (2018). Neural tangent kernel. *NeurIPS.* PS4 A3

[S] Lee, J., Bahri, Y., Novak, R., et al. (2018). Deep neural networks as Gaussian processes. *ICLR.* — With Neal (1996), the infinite-width correspondence.

[S] Neyshabur, B., Tomioka, R., & Srebro, N. (2015). In search of the real inductive bias. *ICLR workshop.* — Early and clear statement of the puzzle.

[S] Frankle, J., & Carbin, M. (2019). The lottery ticket hypothesis. *ICLR.*

[R] Neal, R. M. (1996). *Bayesian Learning for Neural Networks.* Springer. — The original infinite-width Gaussian-process result.

[R] Bartlett, P. L., Foster, D. J., & Telgarsky, M. (2017). Spectrally-normalized margin bounds for neural networks. *NeurIPS.* — The most serious attempt at a norm-based bound.

[R] Chizat, L., Oyallon, E., & Bach, F. (2019). On lazy training in differentiable programming. *NeurIPS.* — Why the NTK regime does not describe practical training.

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6. Convolutional networks

[C] LeCun, Y., Bottou, L., Bengio, Y., & Haffner (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11). §I–II. [LeNet]

[C] Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). ImageNet classification with deep convolutional neural networks. *NeurIPS.* [AlexNet]

[C] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *CVPR.* [ResNet] — The most consequential architectural idea in the course after attention itself; the Transformer inherits it directly. §22 item 15, PS2 A3, **B4**

[C] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Identity mappings in deep residual networks. *ECCV.* — Pre-activation ordering and the clean gradient-path argument. PS2 A3

[S] Simonyan, K., & Zisserman, A. (2015). Very deep convolutional networks. *ICLR.* [VGG]

[S] Szegedy, C., Liu, W., Jia, Y., et al. (2015). Going deeper with convolutions. *CVPR.* [GoogLeNet]

[S] Fukushima, K. (1980). Neocognitron. *Biological Cybernetics*, 36. — Convolution and pooling, 1980.

[S] Zeiler, M. D., & Fergus, R. (2014). Visualizing and understanding convolutional networks. *ECCV*.

[R] Luo, W., Li, Y., Urtasun, R., & Zemel, R. (2016). Understanding the effective receptive field in deep convolutional neural networks. *NeurIPS*. — *New to this revision.* The $O(k\sqrt{L})$ result, as against the $O(kL)$ theoretical field. Lab 4 measures the radius empirically by backpropagating from the centre unit and fitting the exponent, so this is now the reference for a laboratory rather than optional depth. **Lab 4(b,c)**

[R] Huang, G., Liu, Z., van der Maaten, L., & Weinberger, K. (2017). Densely connected convolutional networks. *CVPR*.

[R] Tan, M., & Le, Q. (2019). EfficientNet: rethinking model scaling for CNNs. *ICML*.

[R] Liu, Z., Mao, H., Wu, C.-Y., et al. (2022). A ConvNet for the 2020s. *CVPR*. [ConvNeXt] — What happens when a CNN is given the Transformer's training recipe; useful for separating architecture from recipe.

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7. Recurrent networks and sequence models

[C] Bengio, Y., Simard, P., & Frasconi, P. (1994). Learning long-term dependencies with gradient descent is difficult. *IEEE Trans. Neural Networks*, 5(2). §22 item 16

[C] Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8). §22 item 17, midterm

[C] Pascanu, R., Mikolov, T., & Bengio, Y. (2013). On the difficulty of training recurrent neural networks. *ICML*. — The geometric account and gradient clipping. The proof of the vanishing-gradient bound is examinable. §22 item 16, midterm

[C] Sutskever, I., Vinyals, O., & Le, Q. V. (2014). Sequence to sequence learning with neural networks. *NeurIPS*. **Lab 5**

[C] Cho, K., van Merriënboer, B., Gulcehre, C., et al. (2014). Learning phrase representations using RNN encoder–decoder. *EMNLP*. [GRU] — **§4 especially**, for the length-degradation curves that motivate Week 6. **Lab 5 (all)**

> **Read Cho §4 after Lab 5, not before.** This is the one instruction of its kind in the course. Lab 5 asks you to produce the length-degradation curve yourself, at three context dimensions, and then to say what would fix it — and the entire pedagogical mechanism of Week 5 depends on that being a measurement you made rather than a figure you were shown. If you have already read it, say so in your Lab 5 report. It costs you nothing and it tells us how to read your part (c).

[S] Elman, J. L. (1990). Finding structure in time. *Cognitive Science*, 14(2).

[S] Gers, F. A., Schmidhuber, J., & Cummins, F. (2000). Learning to forget: continual prediction with LSTM. *Neural Computation*, 12(10). — The forget gate is not in the 1997 paper.

[S] **Greff, K., Srivastava, R. K., Koutník, J., et al.** (2017). LSTM: a search space odyssey. *IEEE TNNLS*. — Systematic gate ablation. Most of the complexity is unnecessary.

[S] **Chung, J., Gulcehre, C., Cho, K., & Bengio, Y.** (2014). Empirical evaluation of gated recurrent neural networks.

[R] **Graves, A.** (2013). Generating sequences with recurrent neural networks.

[R] **Karpathy, A., Johnson, J., & Fei-Fei, L.** (2016). Visualizing and understanding recurrent networks. *ICLR workshop*.

[R] **Olah, C.** (2015). Understanding LSTM networks. Blog. — The standard diagram. Read after the derivation, not instead of it.

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8. Word representations and language modeling

[C] **Bengio, Y., Ducharme, R., Vincent, P., & Jauvin, C.** (2003). A neural probabilistic language model. *JMLR*, 3. — Distributed representations and the neural LM, a decade early.

[C] **Mikolov, T., Chen, K., Corrado, G., & Dean, J.** (2013). Efficient estimation of word representations in vector space. [word2vec]

[C] **Sennrich, R., Haddow, B., & Birch, A.** (2016). Neural machine translation of rare words with subword units. *ACL*. [BPE] **PS5 B1**

[S] **Mikolov, T., Sutskever, I., Chen, K., et al.** (2013). Distributed representations of words and phrases and their compositionality. *NeurIPS*. — Negative sampling.

[S] **Pennington, J., Socher, R., & Manning, C. D.** (2014). GloVe. *EMNLP*.

[S] **Levy, O., & Goldberg, Y.** (2014). Neural word embedding as implicit matrix factorization. *NeurIPS*. — Skip-gram with negative sampling factorizes a shifted PMI matrix. Demystifies word2vec.

[S] **Kudo, T., & Richardson, J.** (2018). SentencePiece. *EMNLP*.

[R] **Collobert, R., Weston, J., Bottou, L., et al.** (2011). Natural language processing (almost) from scratch. *JMLR*, 12.

[R] **Bojanowski, P., Grave, E., Joulin, A., & Mikolov, T.** (2017). Enriching word vectors with subword information. *TACL*. [fastText]

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9. Attention and the Transformer

[C] **Bahdanau, D., Cho, K., & Bengio, Y.** (2015). Neural machine translation by jointly learning to align and translate. *ICLR*. — Attention, in the original. Read as the answer to Week 5's measurement. **Lab 6**

[C] Luong, M.-T., Pham, H., & Manning, C. D. (2015). Effective approaches to attention-based neural machine translation. *EMNLP*. — Dot-product scoring; global versus local. **Lab 6(a)**

[C] Vaswani, A., Shazeer, N., Parmar, N., et al. (2017). Attention is all you need. *NeurIPS*. — **Read in full, twice.** Table 1 (path length and complexity) is the architecture's justification; Table 3 (ablations) is the assignment. §22 items 18–21 and 23; PS3 (all); **Lab 7**

[C] Leviathan, Y., Kalman, M., & Matias, Y. (2023). Fast inference from Transformers via speculative decoding. *ICML*. — *Promoted from reference to core in this revision.* Week 14 derives speculative decoding as the response to the batch-size-1 bandwidth diagnosis, and Lab 14 implements the acceptance rule and verifies statistically that the output distribution is exactly the target model's. It cannot remain a reference item when a laboratory depends on its correctness argument. **Lab 14(a,b,c)**

[C] Dosovitskiy, A., Beyer, L., Kolesnikov, A., et al. (2021). An image is worth 16×16 words. *ICLR*. [ViT] — The demonstration that the grid prior was contingent, not necessary.

[S] Xu, K., Ba, J., Kiros, R., et al. (2015). Show, attend and tell. *ICML*. §3–4. — Soft versus hard attention.

[S] Rush, A. (2018). The annotated Transformer. — The paper as executable code. The best companion to PS3 — but note that PS3 forbids `nn.MultiheadAttention` and `nn.Transformer`, so read it for structure and write your own.

[S] Alammar, J. The illustrated Transformer. Blog. — Read after the lecture, for consolidation.

[S] Shaw, P., Uszkoreit, J., & Vaswani, A. (2018). Self-attention with relative position representations. *NAACL*.

[S] Su, J., Lu, Y., Pan, S., et al. (2021). RoFormer: enhanced Transformer with rotary position embedding. [RoPE] — Now near-universal. The derivation showing the inner product depends only on relative position is short and worth doing.

[S] Press, O., Smith, N. A., & Lewis, M. (2022). Train short, test long: attention with linear biases. *ICLR*. [ALiBi]

[S] Jain, S., & Wallace, B. C. (2019). Attention is not explanation. *NAACL*. — With the next entry, the exchange on whether attention weights license interpretive claims.

[S] Wiegreffe, S., & Pinter, Y. (2019). Attention is not not explanation. *EMNLP*.

[S] Elhage, N., Nanda, N., Olsson, C., et al. (2021). A mathematical framework for Transformer circuits. — QK and OV circuits as a decomposition of what attention heads compute.

[S] Olsson, C., Elhage, N., Nanda, N., et al. (2022). In-context learning and induction heads. — The most concrete mechanistic account of where in-context learning comes from.

[C] Dao, T., Fu, D., Ermon, S., Rudra, A., & Ré, C. (2022). FlashAttention. *NeurIPS*. — *Promoted from reference to core in this revision.* Same mathematics, different memory-hierarchy usage. PS4 requires you to measure peak memory and wall-clock across five sequence lengths

and to verify that the FLOP count is identical between the two implementations — which is the entire lesson, and is invisible if you only read the abstract. **PS4 B6**

[R] Dao, T. (2023). FlashAttention-2.

[R] Child, R., Gray, S., Radford, A., & Sutskever, I. (2019). Generating long sequences with sparse Transformers.

[R] Beltagy, I., Peters, M. E., & Cohan, A. (2020). Longformer.

[R] Katharopoulos, A., Vyas, A., Pappas, N., & Fleuret, F. (2020). Transformers are RNNs. *ICML.* — Linear attention, and the equivalence that names the paper.

[R] Choromanski, K., Likhosherstov, V., Dohan, D., et al. (2021). Rethinking attention with Performers. *ICLR.*

[R] Tay, Y., Dehghani, M., Bahri, D., & Metzler, D. (2022). Efficient Transformers: a survey. *ACM Computing Surveys.*

[R] Tay, Y., Dehghani, M., Abnar, S., et al. (2021). Long Range Arena. *ICLR.* — The benchmark on which most efficient-attention claims fail to hold up.

[R] Gu, A., Goel, K., & Ré, C. (2022). Efficiently modeling long sequences with structured state spaces. *ICLR.* [S4]

[R] Gu, A., & Dao, T. (2023). Mamba: linear-time sequence modeling with selective state spaces. — Recurrence returns with a parallel scan. Week 14.

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10. Pretraining, scaling, and foundation models

[C] Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT. *NAACL.*

[C] Brown, T., Mann, B., Ryder, N., et al. (2020). Language models are few-shot learners. *NeurIPS.* [GPT-3]

[C] Kaplan, J., McCandlish, S., Henighan, T., et al. (2020). Scaling laws for neural language models. §1–3. — You are not reading this for its conclusions; you are reading it to find the methodological choice that produced them. See §16. PS5 A2, **Lab 11(c)**

[C] Hoffmann, J., Borgeaud, S., Mensch, A., et al. (2022). Training compute-optimal large language models. *NeurIPS.* [Chinchilla] — The correction to Kaplan. Most models of the preceding era were badly undertrained. §22 item 27, PS5 A1, **Lab 11**

[C] Wei, J., Tay, Y., Bommasani, R., et al. (2022). Emergent abilities of large language models. *TMLR.* PS5 A5

[C] Schaeffer, R., Miranda, B., & Koyejo, S. (2023). Are emergent abilities of large language models a mirage? *NeurIPS.* — Assign with the above. The dispute is about metric discontinuity, and it is unresolved. PS5 A5

[S] Radford, A., Wu, J., Child, R., et al. (2019). Language models are unsupervised multitask learners. [GPT-2] — The default final project implements this.

[S] Raffel, C., Shazeer, N., Roberts, A., et al. (2020). Exploring the limits of transfer learning with a unified text-to-text Transformer. *JMLR*. [T5]

[S] Rajbhandari, S., Rajbhandari, J., Ruwase, O., & He, Y. (2020). ZeRO. *SC20*.

[R] Grattafiori, A., et al. (2024). The Llama 3 herd of models. — The most detailed public account of an end-to-end training pipeline.

[R] Penedo, G., et al. (2024). The FineWeb datasets. — What pretraining data curation actually involves.

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11. Post-training and adaptation

[C] Ouyang, L., Wu, J., Jiang, X., et al. (2022). Training language models to follow instructions with human feedback. *NeurIPS*. [InstructGPT] PS5 B3

[C] Rafailov, R., Sharma, A., Mitchell, E., et al. (2023). Direct preference optimization. *NeurIPS*. [DPO] — The derivation is short and belongs on the exam. §22 item 33, PS5 A3, **B4**

[C] Hu, E. J., Shen, Y., Wallis, P., et al. (2021). LoRA. *ICLR*. **Lab 12 (all)**

[S] Christiano, P., Leike, J., Brown, T., et al. (2017). Deep reinforcement learning from human preferences. *NeurIPS*.

[S] Schulman, J., Wolski, F., Dhariwal, P., et al. (2017). Proximal policy optimization.

[S] Houlsby, N., Giurgiu, A., Jastrzebski, S., et al. (2019). Parameter-efficient transfer learning for NLP. *ICML*. — Adapters.

[S] Wei, J., Wang, X., Schuurmans, D., et al. (2022). Chain-of-thought prompting. *NeurIPS*.

[R] Wang, X., Wei, J., Schuurmans, D., et al. (2023). Self-consistency improves chain-of-thought reasoning. *ICLR*.

[R] Lewis, P., Perez, E., Piktus, A., et al. (2020). Retrieval-augmented generation. *NeurIPS*.

[R] Snell, C., Lee, J., Xu, K., & Kumar, A. (2024). Scaling LLM test-time compute optimally.

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12. Representation learning

[C] Bengio, Y., Courville, A., & Vincent, P. (2013). Representation learning: a review and new perspectives. *IEEE TPAMI*, 35(8).

[C] van den Oord, A., Li, Y., & Vinyals, O. (2018). Representation learning with contrastive predictive coding. [InfoNCE] §22 item 32, **Lab 10(a)**

[C] Chen, T., Kornblith, S., Norouzi, M., & Hinton, G. (2020). A simple framework for contrastive learning. *ICML*. [SimCLR] **Lab 10(b)**

[C] Radford, A., Kim, J. W., Hallacy, C., et al. (2021). Learning transferable visual models from natural language supervision. *ICML*. [CLIP]

[S] He, K., Fan, H., Wu, Y., Xie, S., & Girshick, R. (2020). Momentum contrast. *CVPR*. [MoCo]

[S] He, K., Chen, X., Xie, S., et al. (2022). Masked autoencoders are scalable vision learners. *CVPR*. [MAE]

[S] Grill, J.-B., Strub, F., Altché, F., et al. (2020). Bootstrap your own latent. *NeurIPS*. [BYOL] — Contrastive learning without negatives. Why it does not collapse remains partly unexplained.

[S] Wang, T., & Isola, P. (2020). Understanding contrastive representation learning through alignment and uniformity. *ICML*.

[S] Caron, M., Touvron, H., Misra, I., et al. (2021). Emerging properties in self-supervised vision Transformers. *ICCV*. [DINO]

[R] Kingma, D. P., & Welling, M. (2014). Auto-encoding variational Bayes. *ICLR*. — For the one-lecture generative context.

[R] Goodfellow, I., Pouget-Abadie, J., Mirza, M., et al. (2014). Generative adversarial networks. *NeurIPS*.

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13. Evaluation and practice

[C] Karpathy, A. (2019). A recipe for training neural networks. Blog. — Week 13. The most useful thousand words in the course. *everything*

[C] Papineni, K., Roukos, S., Ward, T., & Zhu, W.-J. (2002). BLEU. *ACL*. — Read for its limitations as much as its method. **Lab 5(a), Lab 13(b)**

[C] Hendrycks, D., Burns, C., Basart, S., et al. (2021). Measuring massive multitask language understanding. *ICLR*. [MMLU] PS5 B5

[C] Liang, P., Bommasani, R., Lee, T., et al. (2022). Holistic evaluation of language models. [HELM] PS5 B5

[S] Zinkevich, M. Rules of machine learning: best practices for ML engineering. — Systems-oriented; complements Karpathy.

[S] Zheng, L., Chiang, W.-L., Sheng, Y., et al. (2023). Judging LLM-as-a-judge. *NeurIPS*. — Position bias, verbosity bias, self-preference.

[S] Goodfellow et al. (2016), Ch. 11: Practical methodology.

[R] Ruder, S. Challenges and opportunities in NLP benchmarking. Blog.

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14. Robustness, distribution shift, interpretability

[S] **Szegedy, C., Zaremba, W., Sutskever, I., et al.** (2014). Intriguing properties of neural networks. *ICLR.* — Adversarial examples.

[S] **Madry, A., Makelov, A., Schmidt, L., et al.** (2018). Towards deep learning models resistant to adversarial attacks. *ICLR.*

[S] **Geirhos, R., Jacobsen, J.-H., Michaelis, C., et al.** (2020). Shortcut learning in deep neural networks. *Nature Machine Intelligence.*

[R] **Koh, P. W., Sagawa, S., Marklund, H., et al.** (2021). WILDS. *ICML.*

[R] **Ilyas, A., Santurkar, S., Tsipras, D., et al.** (2019). Adversarial examples are not bugs, they are features. *NeurIPS.*

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15. Perspective and history

[C] **Hooker, S.** (2021). The hardware lottery. *Communications of the ACM*, 64(12). — Week 14. The frame for asking whether attention won on merit or on hardware fit.

[S] **Sutton, R.** (2019). The bitter lesson. Blog. — Short, contested, worth arguing about.

[S] **LeCun, Y., Bengio, Y., & Hinton, G.** (2015). Deep learning. *Nature*, 521. — The field's own summary at the moment before Transformers.

[R] **Schmidhuber, J.** (2015). Deep learning in neural networks: an overview. *Neural Networks*, 61. — Exhaustive on priority and attribution.

[R] **Manning, C. D.** (2022). Human language understanding and reasoning. *Daedalus.*

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16. The four you are expected to contradict

Most of this list is read to be believed. Four entries are read to be checked, and in each case the assessment expects your number to differ from the published one. **Reporting the disagreement honestly earns more than manufacturing agreement.**

Vaswani et al., Table 3. PS3 assigns you one row, pooled across the cohort, capped at four configurations and three seeds. At Multi30k scale most rows do not reproduce cleanly, and for at least one the effect is smaller than the seed variance. The required output is whether your effect size exceeds that variance — not a number that matches the paper.

Hoffmann et al. Lab 11's pooled fit across eighteen cells on three iso-FLOP contours will not give 20 tokens per parameter. The assignment asks for your two largest sources of error, and for an error bar. Note also that a scale-independent ratio would require the exponent to be exactly 0.50; at 0.45 the ratio drifts with compute, so "20 tokens per parameter" is a value at a scale rather than a constant.

Kaplan et al. You will run Lab 11 on a harness that fixes the cosine schedule length across runs of different duration — the same methodological choice, deliberately reproduced. You are asked which direction it biases the result, why the bias is systematic rather than noise, and whether your own pooled fit shows it. You reproduce the historical error before being shown the correction, which is the only way the correction is memorable.

Grill et al. Lab 10's no-negatives-with-stop-gradient run may or may not collapse for you at 20 epochs. If it does, the required answer is not that BYOL is wrong. It is a statement of what you would need to change to distinguish "negatives are necessary" from "my predictor was misconfigured" — the same skill the Reddi et al. entry in §3 is assigned to build.

Four is not many out of roughly a hundred and forty-five entries. But a reading list on which nothing ever fails to replicate is teaching a literature rather than a science, and these four are where the difference is visible at the scale you can afford.

Appendix — Reading and laboratory load by week

Pages are for the core **[C]** readings only. The laboratory column is the GPU-hour budget from the syllabus, not contact time; every laboratory is 50 minutes at the desk regardless.

Wk	Core [C]	pp	Laboratory	GPU-h
1	Prince 2–6; Kingma & Ba; Loshchilov & Hutter	90	L1 Numerics before networks	0
2	Rumelhart et al.; Baydin et al.; Karpathy	55	L2 The zoo of silent failures	0.2
3	Telgarsky §2,§5; Barron; Glorot & Bengio; He et al.	70	L3 The exponential-failure table	0.2
4	Prince 10–11; LeCun §I–II; Krizhevsky; He $\times 2$; Ioffe & Szegedy; Ba; Santurkar	110	L4 Receptive fields, measured	0.5
5	Bengio '94; Hochreiter & Schmidhuber; Pascanu; Bengio '03; Sennrich; Sutskever; Cho §4 *(after the lab)*	105	L5 **The bottleneck**	2.5

Wk	Core [C]	pp	Laboratory	GPU-h
6	Bahdanau; Luong; Vaswani §1–3	40	L6 Attention, and the payoff	1.5
7	Vaswani (full, ×2); Xiong; Dosovitskiy	45	L7 The parameter count	0.5
8	Zhang et al.; Belkin; Nakkiran; Jacot; Wilson	90	L8 Memorization	2
9	Yang & Hu; Yang et al.; McCandlish; Goyal; Dao	100	L9 Coordinate check, roofline	6
10	Bengio ’13; van den Oord; Chen; Radford	70	L10 Collapse, and preventing it	4
11	Kaplan §1–3; Hoffmann; Wei; Schaeffer; Devlin; Brown	100	L11 Scaling laws on a budget	1
12	Ouyang; Rafailov; Hu; Su	65	L12 LoRA and RoPE	1
13	Karpathy; Papineni; Hendrycks; Liang (skim)	60	L13 The metric is an instrument	0.5
14	Hooker; Snell; Leviathan; Gu & Dao	55	L14 Bandwidth, and cheating it	1

Roughly 60–70 pages of primary literature per week, and roughly 1.5 GPU-hours. Papers are read twice: once before lecture for orientation, once after for detail. Weeks 4, 9 and 11 are the three heavy weeks on both axes at once, and Week 9 is the one to plan around — 100 pages against a reserved overnight cluster window.